

TID	Items Bought
100	<i>f, a, c, d, g, i, m, p</i>
200	<i>a, b, c, f, l, m, o</i>
300	<i>b, f, h, j, o</i>
400	<i>b, c, k, s, p</i>
500	<i>a, f, c, e, l, p, m, n</i>

Minimum support count=3

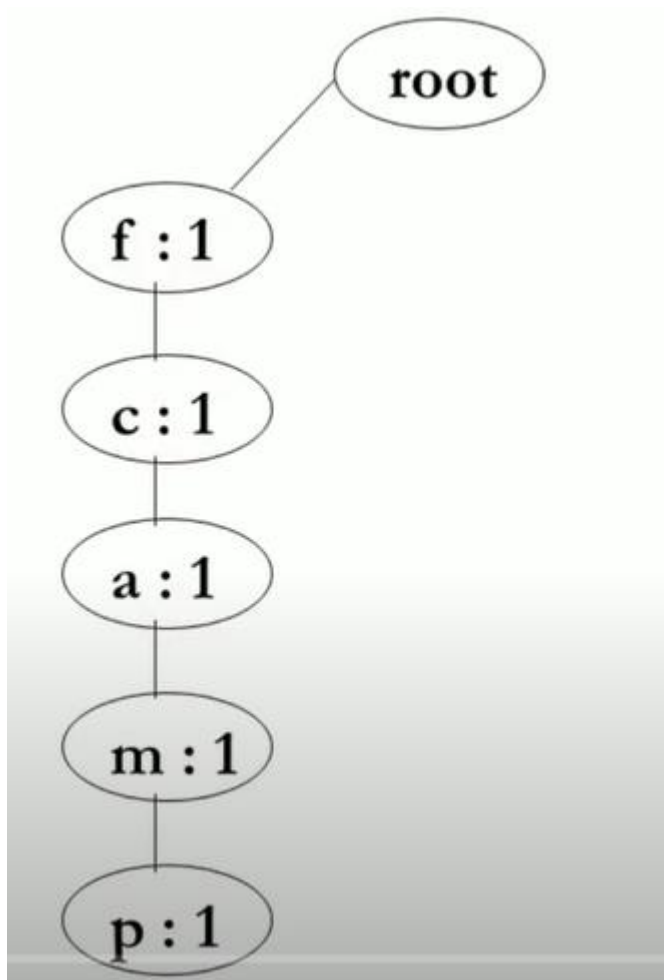
TID	Items Bought
100	<i>f, a, c, d, g, i, m, p</i>
200	<i>a, b, c, f, l, m, o</i>
300	<i>b, f, h, j, o</i>
400	<i>b, c, k, s, p</i>
500	<i>a, f, c, e, l, p, m, n</i>

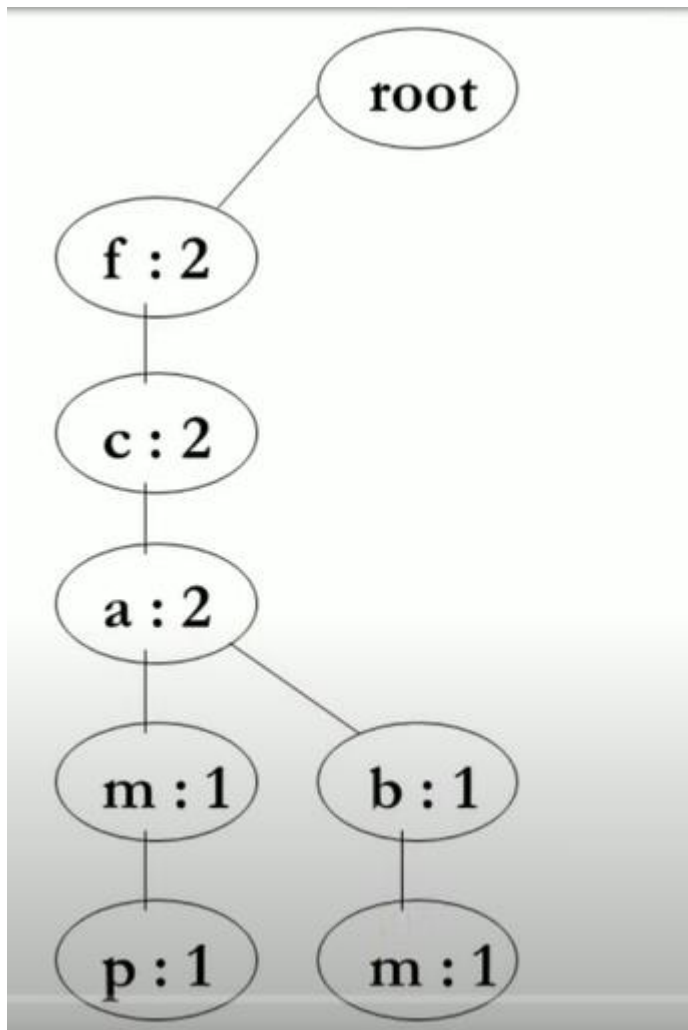
Minimum Support - 3

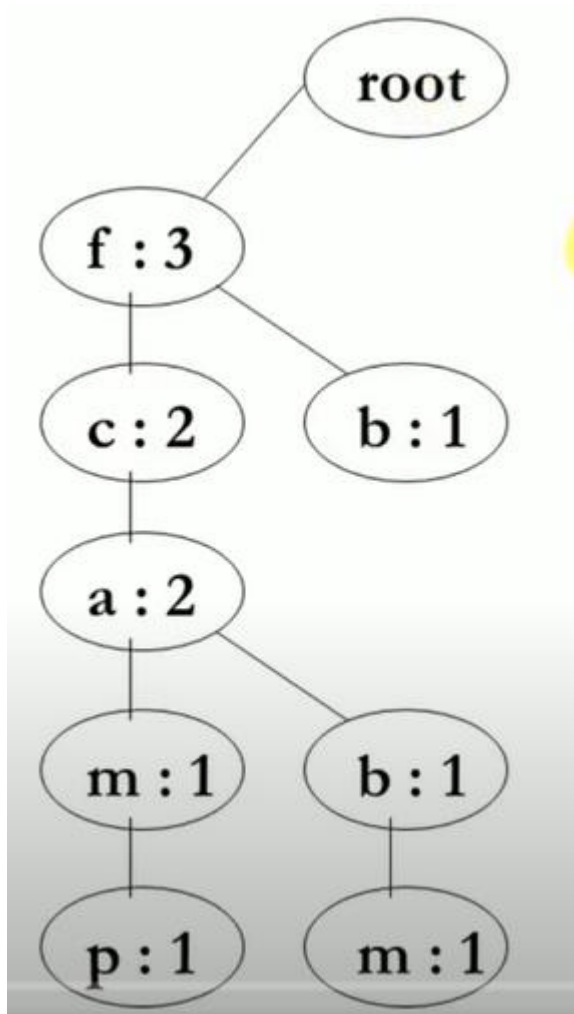
Item	Frequency	Item	Frequency
a	3	j	1
b	3	k	1
c	4	l	2
d	1	m	3
e	1	n	1
f	4	o	2
g	1	p	3
h	1	s	1
i	1		

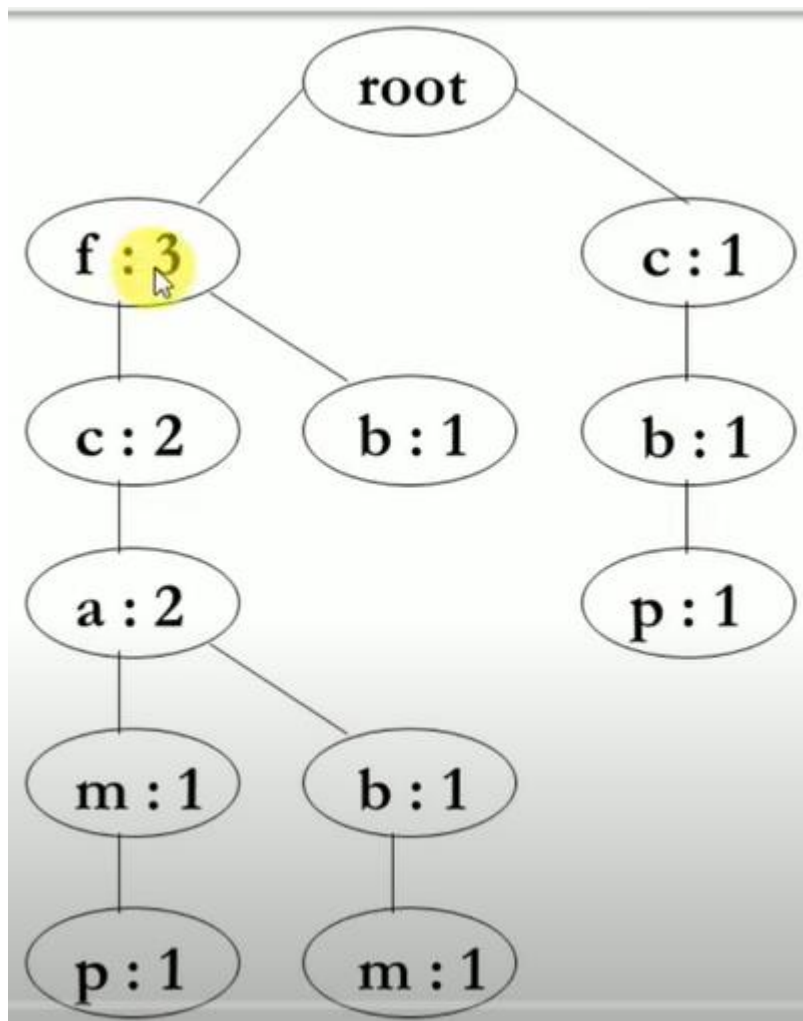
$L = \{ (f:4), (c:4), (a:3), (b:3), (m:3), (p:3) \}$

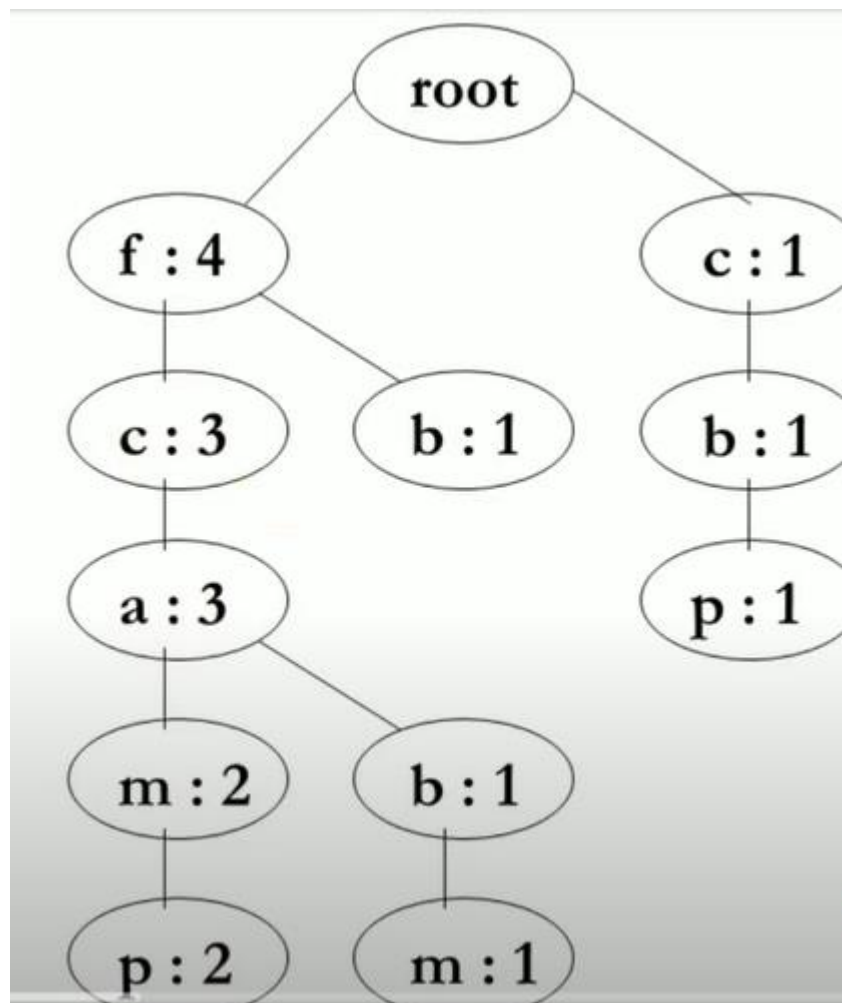
TID	Items Bought	(Ordered) Frequent Items
100	<i>f, a, c, d, g, i, m, p</i>	<i>f, c, a, m, p</i>
200	<i>a, b, c, f, l, m, o</i>	<i>f, c, a, b, m</i>
300	<i>b, f, h, j, o</i>	<i>f, b</i>
400	<i>b, c, k, s, p</i>	<i>c, b, p</i>
500	<i>a, f, c, e, l, p, m, n</i>	<i>f, c, a, m, p</i>











Item	Conditional Pattern Base
p	$\{\{f, c, a, m : 2\}, \{c, b : 1\}\}$
m	$\{\{f, c, a : 2\}, \{f, c, a, b : 1\}\}$
b	$\{\{f, c, a : 1\}, \{f : 1\}, \{c : 1\}\}$
a	$\{\{f, c : 3\}\}$
c	$\{\{f : 3\}\}$
f	Φ

Item	Conditional Pattern Base	Conditional FP-Tree
p	$\{\{f, c, a, m : 2\}, \{c, b : 1\}\}$	$\{c : 3\}$
m	$\{\{f, c, a : 2\}, \{f, c, a, b : 1\}\}$	$\{f, c, a : 3\}$
b	$\{\{f, c, a : 1\}, \{f : 1\}, \{c : 1\}\}$	Φ
a	$\{\{f, c : 3\}\}$	$\{f, c : 3\}$
c	$\{\{f : 3\}\}$	$\{f : 3\}$
f	Φ	Φ

Item	Conditional Pattern Base	Conditional FP-Tree	Frequent Patterns Generated
p	$\{\{f, c, a, m : 2\}, \{c, b : 1\}\}$	$\{c : 3\}$	$\{<c, p : 3>\}$
m	$\{\{f, c, a : 2\}, \{f, c, a, b : 1\}\}$	$\{f, c, a : 3\}$	$\{$ $<f, m : 3>$, $<c, m : 3>$ $<a, m : 3>$, $<f, c, m : 3>$ $<f, a, m : 3>$, $<c, a, m : 3>\}$
b	$\{\{f, c, a : 1\}, \{f : 1\}, \{c : 1\}\}$	Φ	$\{\}$
a	$\{\{f, c : 3\}\}$	$\{f, c : 3\}$	$\{<f, a : 3>, <c, a : 3>, <f, c, a : 3>\}$
c	$\{\{f : 3\}\}$	$\{f : 3\}$	$\{<f, c : 3>\}$
f	Φ	Φ	$\{\}$

Apriori Algorithm-example:

Transactions List

1	Milk	Egg	Bread	Butter
2	Milk	Butter	Egg	Ketchup
3	Bread	Butter	Ketchup	
4	Milk	Bread	Butter	
5	Bread	Butter	Cookies	
6	Milk	Bread	Butter	Cookies
7	Milk	Cookies		
8	Milk	Bread	Butter	
9	Bread	Butter	Egg	Cookies
10	Milk	Butter	Bread	
11	Milk	Bread	Butter	
12	Milk	Bread	Cookies	Ketchup

Minimum support=33%

Minimum confidence=50%

1-item Sets	Frequency
Milk	9
Bread	10
Butter	10
Egg	3
Ketchup	3
Cookies	5

Frequent 1-item Sets	Frequency
Milk	9
Bread	10
Butter	10
Cookies	5

2-item Sets	Frequency
Milk, Bread	7
Milk, Butter	7
Milk, Cookies	3
Bread, Butter	9
Butter, Cookies	3
Bread, Cookies	4

Frequent 2-item Sets	Frequency
Milk, Bread	7
Milk, Butter	7
Bread, Butter	9
Bread, Cookies	4

3-item Sets	Frequency
Milk, Bread, Butter	6
Milk, Bread, Cookies	1
Bread, Butter, Cookies	3
Milk, Butter, Cookies	2

Frequent 3-item Sets	Frequency
Milk, Bread, Butter	6

- Frequent 3-Item Set = $I \Rightarrow \{Milk, Bread, Butter\}$
- Non-Empty subset are
 - $\{\{Milk\}, \{Bread\}, \{Butter\}, \{Milk, Bread\}, \{Milk, Butter\}, \{Bread, Butter\}\}$

- Rule 1: {Milk} → {Bread, Butter} {S=50%, C=66.67%}
 - Support = $6/12 = 50\%$
 - Confidence = $\text{Support (Milk, Bread, Butter)} / \text{Support(Milk)} = \frac{6/12}{9/12} = 6/9 = 66.67\% > 60\%$
 - Valid

- Rule 2: {Bread} → {Milk, Butter} {S=50%, C=60%}
 - Support = $6/12 = 50\%$
 - Confidence = $\text{Support (Milk, Bread, Butter)} / \text{Support(Bread)} = 6/10 = 60\% \geq 60\%$
 - Valid

- Rule 3: {Butter} → {Milk, Bread} {S=50%, C=60%}
 - Support = $6/12 = 50\%$
 - Confidence = $\text{Support (Milk, Bread, Butter)} / \text{Support(Butter)} = 6/10 = 60\% \geq 60\%$
 - Valid

- Rule 4: {Milk, Bread} → {Butter} {S=50%, C=85.7%}
 - Support = $6/12 = 50\%$
 - Confidence = $\text{Support (Milk, Bread, Butter)} / \text{Support(Milk, Bread)} = 6/7 = 85.7\% > 60\%$
 - Valid

- Rule 5: {Milk, Butter} → {Bread} {S=50%, C=85.7%}
 - Support = $6/12 = 50\%$
 - Confidence = $\text{Support (Milk, Bread, Butter)} / \text{Support(Milk, Butter)} = 6/7 = 85.7\% \geq 60\%$
 - Valid

- Rule 6: {Bread, Butter} → {Milk} {S=50%, C=66.67%}
 - Support = $6/12 = 50\%$
 - Confidence = $\text{Support (Milk, Bread, Butter)} / \text{Support(Bread, Butter)} = 6/9 = 66.67\% \geq 60\%$
 - Valid